

**Greater Birmingham &
Solihull LEP**

West Midlands Creative Business scale-up mapping

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Executive summary

West Midlands Combined Authority (WMCA) is one of three city regions to have agreed a Sector Deal for Creative Industries. This includes funding for a creative business scale-up programme, to boost growth and productivity in the sector across all parts of the region.

WMCA needed to establish which creative businesses in the region would most likely be eligible for, and derive most benefit from, this programme. The Common Application Gateway (CAG) criteria established for the programme by DCMS – turnover growth of 10% and a minimum of 8 employees (a mixture of four permanent and four freelance workers) – represent a high bar for the majority of West Midlands creative businesses. A more detailed and specialist account of the creative sector was needed to unearth the latent scale-up potential of creative businesses across the city region.

BOP had previously carried out a mapping of the creative economy in Greater Birmingham & Solihull in 2017/18, using Business Register and Employment Survey (BRES) data analysed by standard DCMS Creative Industries sector definitions to compile a dataset of creative businesses. This was analysed for the strengths and opportunities both within the creative industries sub-sectors defined by DCMS¹ and across four supply chains defined by product type:

- Creative Content: eg, television, film, digital media, software and applications, computer games, publishing, recorded music
- Creative Experiences: eg, live performance, venues, museums, galleries and libraries
- Creative Originals: eg, visual arts, fashion, crafts, product design
- Creative Services: eg, advertising and marketing, architecture crafts, graphic design, design consultancy, computer services

For this study, BOP joined with City Curator to pursue an innovative and multi-method approach to mapping creative businesses and workforce to provide both a fine-grained understanding of creative businesses' 'scale-up potential' and contact, financial and employment details of individual creative businesses.

Initial mapping of the scale and significance of the sector in September 2019 identified some 14,900 businesses registered at Companies House under one of the DCMS Creative Industries subsectors with an office in one of the WMCA constituent and non-constituent member and partner areas.

Initial analysis of this cohort to establish details of employment, turnover, profitability and assets resulted in a 'longlist' of 2,336 businesses for further analysis by a range of methods including AI web search, social media analysis, records of public grants and proprietary investment databases. The method used LinkedIn data to provide more up to date information on company size, location and activities. This resulted in a shortlist of 645 companies, including 49 'non-creative industries businesses' (ie, those not in a DCMS CI SIC code), uncovered through keyword search, whose products or services either clearly formed part of a creative industries supply chain or made extensive use of creative and immersive content and technologies .

As suspected at the outset, very few of the shortlist – 26 companies – met both the CAG criteria, with a further 386 meeting one or other criterion. We applied other measures of private sector investment value, public investment, social media and recruitment activity to establish a level of confidence in the scale-up potential of the shortlisted companies. This indicated that there were 168 West Midlands creative businesses with a very strong likelihood of scale-up potential, and a further 253 which we think could benefit from the programme.

We organised our findings into a colour-coded dashboard (provided as an accompanying document to this report) which also sets out further details of company founders, employment numbers and contact details. The result is a comprehensive toolkit to assist the programme director in marketing and recruitment.

¹ Advertising and marketing; architecture; craft; design; film, television, video, radio and photography; IT, software and computer services; publishing; museums, galleries and libraries; and music, performing and visual arts.

1. Mapping the creative sector

1.1 Total population of creative businesses

Our initial analysis of DueDil identified 14,903 active companies of all ages with West Midlands addresses registered under one or more of the DCMS Creative Industries SIC codes at Companies House. Table 1 **Error! Reference source not found.** shows those companies broken down by age since incorporation.

Table 1 Active DCMS Creative Industries companies within West Midlands

Companies within West Midlands	Number of companies
DCMS Creative Industries SIC code	14,903
DCMS CI SIC and aged less than two years	4,267
DCMS CI SIC and aged 2-10 years	7,291
DCMS CI SIC and aged 11 years or more	3,345

Source: BOP/City Curator analysis of DueDil

1.2 Common application gateway criteria

We tested all 14,903 eligible creative businesses against DCMS's 'common application gateway' (CAG) criteria for the creative business scale-up programme:

- Turnover growth equal to or greater than 10% in the last year (see 1.3, below)
- Eight or more employees (see 1.4, below)

Only a tiny minority of company records - ten companies, of which two were part of the same group – contained data which indicated that the business met both CAG criteria. Below we explain why this information is not available in most cases, and the steps we took to try to mitigate that.

1.3 Measuring turnover growth

Most companies are below the £10.2 million per annum turnover threshold at which they have to submit full accounts to Companies House. This means that most companies submit accounts in one of two forms:

Micro-entity accounts

Companies may submit 'micro-entity accounts' if they meet two of the following criteria:

- Turnover is no more than £632,000
- Balance sheet total is no more than £316,000
- Average number of employees is no more than 10

The minimum level of disclosure for micro-entity accounts consist of:

- Simple balance sheet and footnotes
- Signature of a director and their name printed on the balance sheet.
- Statement on the balance sheet above the director's signature that the accounts have been prepared
- Simple profit and loss account

Whilst companies that can be defined as micro-entities are potentially in scope of the scale-up programme, these limited reporting requirements mean that it is very unlikely that the public record will contain details of turnover or turnover growth.

Abridged accounts

Companies may submit abridged accounts whose:

- Turnover is no more than £10.2 million
- Balance sheet total is no more than £5.1 million
- Average number of employees is no more than 50

Abridged accounts include abbreviated versions of the company balance sheet and profit and loss account. Again, this can mean that the public record does not contain details of turnover; but we are more likely to be able to find evidence of turnover or growth, either year-on-year or on a Compound Annual Growth Rate (CAGR) basis, for these companies.

1.4 Measuring employment

DCMS had already indicated that there was some flexibility around the definition of 'employees'. To reflect the known prevalence of freelance and contract working, the common application gateway criteria could be interpreted to mean businesses with four permanent and four freelance employees in the last year.

However, in most cases it was not possible to determine an exact number of employees from publicly available data, nor a split of permanent and freelance workers. Our main sources of data for employment numbers were:

- Team pages on the company web site – although many companies do not have either a web site or a team page
- Information about the company and its employees from LinkedIn.

LinkedIn now has 25 million UK users and provides a much larger and more accurate sample of employment than BRES data. LinkedIn also generates an estimate of employee numbers based on the number of subscribers with links to the company, for which the lowest band is companies with 2-10 employees. This led us to determine that we would carry out further analysis on all companies for which we could find evidence of at least two employees. This 'permissive' approach allowed us to include more businesses in our initial longlist for further detailed analysis.

1.5 Additional growth criteria

We developed a wider set of growth criteria, based on other financial and employment information available from DueDil, to enlarge the pool of potentially eligible businesses within four areas: scale, growth, profitability and assets. The

list of the eleven additional growth criteria applied are shown in Table 2. **Error! Reference source not found.**

Table 2 Additional growth criteria

Criteria	Definition	Justification
Employees	Two or more employees	Excludes sole traders, which are unlikely to be ready to scale
Employee Growth	25%+ in the last year	Identifies smaller companies whose workforce is growing
Employee Growth	25%+ Compound Annual Growth Rate (CAGR) over the last 3 years	Identifies long term growth; accounts for temporary variations in staffing numbers in a single year
Turnover	8%+ CAGR over 3 years	Identifies long term growth; accounts for temporary variations in turnover in a single year
Turnover Delta	10,000 or greater	An indicator of growth based on the difference in turnover between two financial years.
Pre-tax profit	Positive	Identifies companies that do not report other information or fall below CAG thresholds but are worth further investigation
Pre-tax profit growth	Positive	<i>ibid</i>
Current assets	£100,000 or more	Identifies companies that do not report other financial figures but may be worthy of further investigation
Intangible Assets	£80,000 or more	Identifies companies that do not report other indicators but may be growing in less obvious ways.
Intangible Asset growth	25% +	<i>ibid</i>

2. Selection of the shortlist

2.1 In-depth analysis of the longlist

Our initial analysis has resulted in a cohort of 2,336 businesses – 2,082 aged between two and ten years, 66 younger than two years and 188 between 11 and 15 years old. We carried out further detailed analysis of their scale-up potential to arrive at a 'scale-up score' for each company.

These analyses included:

- **AI web search** that looked for additional evidence of growth factors, eg, by searching for team, project and client pages on the company web site. This:
 - Provided more information about company size, products and services)
 - Helped us categorise businesses into product or services companies, and allocate them to one of four creative supply chains (or none)
 - Gave insight into the scope of their activity for further (manual) analysis
 - Eliminated those companies without a web site
- **Crunchbase**, to establish evidence of private investment within the last three years
- **LinkedIn** data, to provide more accurate estimates of total employment, including freelancers, and details of directors active in the business
- UK Research & Innovation's **Grants for R&D (GRAND)** database, for evidence of successful applications for public funding
- **Recruitment** web sites, to identify companies actively hiring

- **Social media** analysis, to provide further evidence of other growth factors, eg, export activity, exhibitions, new products and services

This returned a shortlist of 645 businesses: 596 of which were registered at Companies House under one of the DCMS CI SIC codes² and 49 of which were registered under a non-DCMS Creative Industries SIC code.

Table 3 Subsector breakdown of shortlisted businesses

DCMS Creative Industries sector	No. of businesses	% of total
Advertising and marketing	79	12%
Architecture	46	7%
Crafts	12	2%
Design: product, graphic and fashion design	48	7%
Film, TV, video, radio and photography	45	7%
IT, software and computer services	240	37%
Publishing	36	6%
Museums, galleries and libraries	1	0%
Music, performing and visual arts	89	14%
Non-DCMS sectors	49	8%
Total	645	100%

Source: BOP Consulting / City Curator analysis 2019

² The Department of Digital, Culture, Media & Skills (DCMS) classifies 'Creative Industries' as consisting of businesses registered within 31 Standard Industrial Classifications (SIC codes), last updated in 2007. It then sub-divided these into nine creative subsectors. See Appendix 2 for a list of SIC codes and sub-sector groupings.

'Non-DCMS' sector businesses fell into two camps:

- Businesses registered under a non-DCMS Creative Industries SIC code whose primary business activity was clearly identified as forming part of creative industries – eg, a film company registered under 'Business support services'
- Business which, whilst working in a market vertical other than creative industries, had a product or service that was wholly or largely dependent on creative or immersive technology (eg, manufacturers of fairground rides and training companies using VR and other immersive technologies)

findings is attached. Further analysis of these numbers is shown in Section 3 below.

2.2 Eligibility

'Eligible' businesses are those which met both the Common Application Gateway criteria established by (DCMS) for admission to the creative business scale-up programme. From the shortlist, only 26 companies met both criteria.

2.3 Confidence (scale-up criteria)

The research methodology was designed to address the risk (crystallised through analysis) that too few West Midlands creative businesses would meet the CAG criteria. The combination of sources detailed at 2.1, above, was used to determine a further four scale-up criteria:

- Private Sector Investment Value
- Public Investment
- Social Media Test
- Recruitment

2.4 Summary of findings

We analysed a shortlist of 645 businesses against the Common Application Gateway criteria and the four scale-up criteria. A summary 'dashboard' of our

3. Eligibility

3.1 Businesses meeting both Common Application Gateway (CAG) Criteria

26 businesses (out of the shortlist of 645) met both CAG criteria, of which 50% were in IT, Software and Computer Services.

Table 4 Companies meeting both Common Application Gateway criteria

DCMS Creative Industries sector	No. of businesses	% of total
Advertising and marketing	0	0%
Architecture	1	4%
Crafts	0	0%
Design: product, graphic and fashion design	0	0%
Film, TV, video, radio and photography	2	8%
IT, software and computer services	13	50%
Publishing	0	0%
Museums, galleries and libraries	0	0%
Music, performing and visual arts	4	15%
Non-DCMS sectors	6	23%
Total	26	100%

Source: BOP Consulting / City Curator analysis 2019

3.2 Businesses meeting one of the CAG criteria

An additional 386 businesses (out of the shortlist of 645) met one or other of the CAG criteria, taking into account a more generous interpretation of company employment which counted both permanent and freelance workers.

Table 5 Companies meeting one of the CAG criteria

DCMS Creative Industries sector	No. of businesses	% of total
Advertising and marketing	55	14%
Architecture	29	8%
Crafts	5	1%
Design: product, graphic and fashion design	30	8%
Film, TV, video, radio and photography	21	5%
IT, software and computer services	151	39%
Publishing	20	5%
Museums, galleries and libraries	0	0%
Music, performing and visual arts	46	12%
Non-DCMS sectors	29	8%
Total	386	100%

Source: BOP Consulting / City Curator analysis 2019

4. Scale-up scores

The following tables summarise our analysis of the 645 businesses using the scale-up scoring methodology described in our proposal.

The scale-up criteria were very useful in discerning the differences between the 619 businesses that did not meet both of the CAG criteria, and yielded the following results:

- None of the shortlisted businesses met all four scale-up criteria (outlined at 2.3 above.)
- 168 businesses met two or three criteria (scored 'Very Good')
- A further 253 businesses met at least one criterion (scored 'Good')
- 224 businesses met none of the investment or recruitment tests and had either no or partial social media coverage (scored 'Low'.)

For the purposes of interpretation, we have assumed that companies scoring 'Very Good' or 'Good' have some scale-up potential. We have not analysed the companies scoring 'Low' in the same level of detail. However, should you wish to pursue those companies further, the report and appendices include data on all 645 shortlisted companies.

Table 6 Businesses meeting two or three of the scale-up criteria

DCMS Creative Industries sector	No. of businesses	% of total
Advertising and marketing	16	10%
Architecture	7	4%
Crafts	1	1%
Design: product, graphic and fashion design	11	7%
Film, TV, video, radio and photography	4	2%
IT, software and computer services	81	48%
Publishing	7	4%
Museums, galleries and libraries	0	0%
Music, performing and visual arts	9	5%
Non-DCMS sectors	32	19%
Total	168	100%

Source: BOP Consulting / City Curator analysis 2019

Table 7 Businesses meeting one scale-up criterion

DCMS Creative Industries sector	No. of businesses	% of total
Advertising and marketing	45	18%
Architecture	13	5%
Crafts	4	2%
Design: product, graphic and fashion design	18	7%
Film, TV, video, radio and photography	17	7%
IT, software and computer services	78	31%
Publishing	15	6%
Museums, galleries and libraries	0	0%
Music, performing and visual arts	55	22%
Non-DCMS sectors	8	3%
Total	253	100%

Source: BOP Consulting / City Curator analysis 2019

5. Correlation between CAG and scale-up scores

We looked at the way that the scale-up scores reflected compliance with CAG criteria. This showed that:

- All businesses meeting both CAG criteria also scored well against scale-up criteria (14 'Very Good' and 12 'Good')
- 297 businesses which met one of the CAG criteria also scored well against scale-up criteria – 129 of which were 'Very Good', and the remaining 168 'Good'
- 98 businesses which failed to meet either CAG criteria showed indications of scale-up potential, with 25 of these scoring 'Very Good' on scale-up confidence

Table 8 Correlation of business performance against CAG criteria and scale-up confidence scores

CAG score	'Very Good' # of companies	'Very Good' % of companies	'Good' # of companies	'Good' % of companies
Pass	14	8%	12	5%
Amber	129	77%	168	66%
Fail	25	15%	73	29%
Total	168	100%	253	100%

Source: BOP Consulting / City Curator analysis 2019

6. Creative supply chain groupings

We further analysed of the results using the methodology established in our 2017/18 mapping to the creative economy of the Greater Birmingham & Solihull LEP area³. This recombined SIC codes of our shortlisted companies into descriptions of creative supply chains established in our previous study.

The 645 shortlisted companies divided into the following creative supply chain groupings:

- Creative Content: 325 (50%)
- Creative Experiences: 50 (8%)
- Creative Originals 57 (9%)
- Creative Services: 183 (28%)

A further 30 businesses (5% of the total shortlist) were included because of their use of creative and immersive technologies in non-creative applications, although they did not fit the definition of any of the four creative supply chains.

³ BOP Consulting (2018). Mapping the Creative Economy of Greater Birmingham & Solihull LEP.

Appendix 1 : Methodology

A range of different data sources were analysed and combined to establish eligibility and scale-up scores. Their use is explained in Table 9, below.

Dashboard

Table 9 Explanation of data sources used in the dashboard

Column header	Description	Source(s)
Company Name	As registered on Companies House	DueDil
Contact Email	Primary contact Companies only contactable via an online form are marked Form.	Company website, DueDil
Contact Name & Profile	Where possible the name and LinkedIn profile of a senior member of staff or the owner has been provided.	Company website, LinkedIn
Website	Several companies use different domain names than the company name	DueDil, web search, LinkedIn
Telephone	Main contact number	DueDil, company website
Subsector	Based on SIC 2007 codes	DueDil
Turnover %	Percentage growth in turnover in the last year for companies reporting this figure	DueDil
Number of Employees	The higher of reported values from company website (eg, the 'Team' or 'About Us' page) or LinkedIn ⁴	Company website, LinkedIn
Private Sector Investment Value	Taken from Crunchbase https://www.crunchbase.com	Crunchbase
Social Media Test	An analysis of the company's social media marketing based on: official company presence on one or more platforms, active profiles (i.e., new activity within the last six months), frequency of posts, reach and integration into other marketing assets	Facebook, Instagram, Twitter, YouTube
Recruitment	Evidence of recent recruitment advertising (0-6 months)	Company website; recruitment websites
Eligibility	Common Application Gateway criteria for scale-up programme (10% turnover growth, 8 employees)	DueDil, LinkedIn

⁴ Note that all companies fit the SME criteria on DueDil but due to the way LinkedIn associates people with companies some of the companies show an employee count larger than the SME maximum. Getting an accurate count for the individual company is impossible without sifting through every single employee

Data

The 'Data' spreadsheet provided as a counterpart to this report includes a larger number of data fields, summarised in Table 10, below

Table 10 Data sources

Column header	Detail / source
Registered company name	DueDil
Registered Address	DueDil
Registered address postcode	DueDil
Age	DueDil
LinkedIn Contact Link	Where possible the name and LinkedIn profile of a senior member of staff or the owner has been provided; the LinkedIn url is hidden
Website	Note that for a variety of reasons several companies use different domain names than the company name
Telephone	From official Companies House records or from website
Subsector	DCMS Creative Industries subsector, or non-DCMS primary SIC code
SIC 2007 Definition	From ONS
Product	Whether the company website identifies a product
Employees on LinkedIn	Number of individuals who show employment at this company on their personal profile
Website employees	The number of employees taken from the company website or social media
Recruitment	Number of jobs advertised in the last six months, obtained from company websites and recruitment sites
Max Employees	Higher of the employment count from the company website or LinkedIn

LinkedIn Company Size	LinkedIn provides an estimate of company size, by size bands: this is initially set by the user but over time LinkedIn analyses collects data on past employees and/or freelancers to improve this score
Turnover Growth (Annual)	Taken from last accounts available on DueDil
Turnover Growth (3 Year CAGR)	Taken from last accounts available on DueDil
Private Funding Value	Taken from Crunchbase: where the value of deals is not disclosed, these have been tagged as 'Unknown'
Public Funding Awards	Number of grants awarded to the company
Public Funding Value	Value of grants awarded: note that a number of these are partnership awards and it is impossible to split out a definitive figure from publicly available data; in these case a rounded figure has been given (eg, >£1m)
Social Media Pass/Fail	Does the company have a social media presence
LinkedIn page	Link to the companies LinkedIn page where available
Facebook	Link to the companies Facebook page where available
Twitter	Link to the companies Twitter feed where available
Insta	Link to the companies Instagram feed where available
YouTube	Link to the companies YouTube channel where available
Ultimate Parent Name	The ultimate parent of the company listed
Geography Check1	A number of companies report an office location in their online profiles to their registered / trading addresses on Companies House. A small number of these appear to be the activity has moved away from the West Midlands. A much larger number are companies that advertise their location as London, but all the indicators suggest that their activity is still based in the region. Companies that require further checks have been flagged.
LinkedIn About Us Abridged	A short (250 word) extract from the companies LinkedIn page – where available

Source: BOP Consulting / City Curator analysis 2019

Appendix 2: DCMS Creative Industries SIC Codes 2015

Table 11 DCMS Creative Industries SIC code groupings

Creative Industries Group	SIC 2007	Description
Advertising and marketing	70.21	Public relations and communication activities
	73.11	Advertising agencies
	73.12	Media representation
Architecture	71.11	Architectural activities
Crafts	32.12	Manufacture of jewellery and related articles
Design: product, graphic and fashion design	74.1	Specialised design activities
Film, TV, video, radio and photography	59.11	Motion picture, video and television programme production activities
	59.12	Motion picture, video and television programme post-production
	59.13	Motion picture, video and television programme distribution
	60.1	Radio broadcasting
	60.2	Television programming and broadcasting activities
	74.2	Photographic activities

IT, software and computer services	58.21	Publishing of computer games
	58.29	Other software publishing
	62.01	Computer programming activities
Publishing	62.02	Computer consultancy activities
	58.11	Book publishing
	58.12	Publishing of directories and mailing lists
	58.13	Publishing of newspapers
	58.14	Publishing of journals and periodicals
	58.19	Other publishing activities
Museums, galleries and libraries	74.3	Translation and interpretation activities
	91.01	Library and archive activities
	91.02	Museum activities
Music, performing and visual arts	59.2	Sound recording and music publishing activities
	85.52	Cultural education
	90.01	Performing arts
	90.02	Support activities to performing arts
	90.03	Artistic creation
	90.04	Operation of arts facilities

Source: DCMS Sectors Economic Estimates Methodology <https://www.gov.uk/government/publications/dcms-sectors-economic-estimates-methodology>

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